



METABOLIC & GLP-1

Tirzepatide

Tirzepatide – 20 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Tirzepatide
SYNONYMS / ALIASES	LY3298176; GIP/GLP-1 receptor dual agonist
CAS NUMBER	2023788-19-2
MOLECULAR FORMULA	C225H348N48O68
MOLECULAR WEIGHT	4813.5 g/mol
AMINO ACID SEQUENCE	YXEGTFTSDYSIXLDKIAQKAFVQWLIAGGPSSGAPPPS (X = alpha-aminoisobutyric acid / Aib at positions 2 and 13; Lys at position 20 is conjugated to a C20 fatty diacid via a gamma-Glu-(2xOEG) linker; C-terminal amide)
SOURCE	Synthetic (solid-phase peptide synthesis with side-chain lipidation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	20 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Lipidated peptides are sensitive to aggregation at the air-water interface - handle with minimal agitation.

STRUCTURAL NOTES

Tirzepatide is a 39-amino-acid synthetic peptide engineered as a dual agonist of the glucose-dependent insulintropic polypeptide (GIP) receptor and the glucagon-like peptide-1 (GLP-1) receptor. Its primary sequence is based on the native GIP backbone with substitutions at positions 2 and 13 (replacing alanine and serine with alpha-aminoisobutyric acid) to confer resistance to dipeptidyl peptidase-4 (DPP-4) cleavage. A C20 fatty diacid is attached to the epsilon-amino group of Lys20 via a gamma-Glu-2x(OEG) linker, enabling reversible albumin binding and a prolonged half-life characteristic of long-acting incretin analogs.

SELECTED PEER-REVIEWED REFERENCES

1. Coskun T. et al. - *LY3298176, a novel dual GIP and GLP-1 receptor agonist for the treatment of type 2 diabetes mellitus: From discovery to clinical proof of concept* - Molecular Metabolism, 2018 - DOI: 10.1016/j.molmet.2018.09.009
2. Frias J.P. et al. - *Tirzepatide versus semaglutide once weekly in patients with type 2 diabetes* - New England Journal of Medicine, 2021 - DOI: 10.1056/NEJMoa2107519
3. Jastreboff A.M. et al. - *Tirzepatide once weekly for the treatment of obesity* - New England Journal of Medicine, 2022 - DOI: 10.1056/NEJMoa2206038

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.